

Yuki Mori

Senior AI Engineer

 yukimori.dev@gmail.com

 Tokyo, Japan

 linkedin.com/in/yukimori-11b56938a

Summary

Senior ML Engineer focused on Retrieval-Augmented Generation (RAG), NLP, and workflow automation (n8n). Built production-grade data pipelines that combine retrieval, reasoning, and automation to deliver measurable business impact. Deep experience with LLMs, vector databases, hybrid retrieval/reranking, prompt engineering, and end-to-end orchestration on cloud. Looking to drive applied AI systems that improve accuracy, reduce latency, and cut manual ops.

Experience

10/2022 – Present
Germany

AI developer

Digital Hero

- Built and deployed production RAG pipelines using LLMs with FAISS/Pinecone and hybrid BM25 + vector + reranking retrieval to improve contextual accuracy.
- Orchestrated n8n workflows for ingestion, embedding, retrieval, and inference with retries, backoffs, and alerting for reliable automation.
- Delivered NLP capabilities including NER, intent detection, and summarization with iterative prompt and query tuning.
- Set up evaluation and observability systems with golden sets, version tracking, and cost/performance dashboards for continuous optimization.
- Hardened production through CI/CD pipelines, PII redaction, and runtime guardrails, achieving 99.95% uptime and <0.2% error rate.

11/2019 – 05/2022
Hong Kong

AI developer

Bilby

- Shipped production Retrieval-Augmented Generation pipelines combining LLMs with FAISS/Pinecone and hybrid BM25 + vector retrieval enhanced by cross-encoder reranking.
- Built and maintained n8n automations for ingestion, embeddings, retrieval, and inference with retries, backoff, and alerting for high reliability.
- Tuned prompts, retrieval parameters, and query-rewriting logic to strengthen precision, recall, and throughput across large-scale evaluations.
- Established robust evaluation and observability frameworks with golden sets, A/B testing, and cost/performance dashboards for continuous optimization.
- Implemented production guardrails and compliance controls with CI/CD integration, ensuring 99.95% uptime and <0.20% error rate across services.

03/2018 – 03/2019
Hong Kong

Internship AI developer

Narrates Limited

- Assisted in developing RAG prototypes that combined LLMs with vector databases (e.g., FAISS/Pinecone) to deliver context-aware internal search and more reliable answers.
- Built n8n workflows to automate data ingestion, retrieval, and inference-with retries, backoff, and alerting-to reduce manual steps and accelerate demo readiness.
- Tuned prompts and retrieval parameters and executed targeted evaluations (golden sets/spot checks) to improve relevance, stability, and consistency across queries.

Projects

05/2022 – 02/2023

Merlin

Digital Hero

- Implemented backend services in Node.js/Express to orchestrate multiple LLM engines and enforce user subscription/entitlement logic.
- Integrated real-time search and summarization: on any visited page, Merlin extracts context, calls model APIs, generates summaries/content, and renders results in an in-page overlay.
- Ensured security and privacy compliance by implementing robust authentication, OAuth flows, encryption at rest/in transit, and aligning with GDPR and internal security standards.

08/2020 – 11/2021

AI voice generator & Text to speech AI voice platform

- Built Node.js/Express backend to orchestrate TTS engines, multi-voice synthesis, and subscription enforcement, integrating Pinecone/FAISS for voice search and Hugging Face models for NLP-driven pronunciation and emotion styling
- Developed real-time text processing pipeline with PyTorch and React/Next.js, enabling instant audio generation, SSML customization, and preview overlays via AWS/Kubernetes-deployed APIs with Redis caching and PostgreSQL/Chroma storage.
- Ensured GDPR compliance with OAuth authentication, encrypted audio storage, and secure voice cloning workflows, supporting 40+ languages and cross-language dubbing.

Skills

RAG & LLMs

Hybrid retrieval/reranking, FAISS/Pinecone, prompt engineering, grounding, knowledge integration

Automation (n8n)

Workflow orchestration, webhooks & APIs, ingestion & embedding & retrieval & inference

Cloud & MLOps

AWS/GCP, Docker, Kubernetes, CI/CD, Redis caching, monitoring

Data

ETL/ingestion, PostgreSQL, MongoDB, Pinecone, Chroma, MySQL, Supabase, Firebase

Programming

Python (PyTorch, NumPy, Pandas), Node.js/Express, React/Next.js/Vite

NLP

NER, intent classification, summarization, Hugging Face/Transformers, offline/online evaluation

Education

09/2016 – 08/2019
Lodz in Poland

Bachelor of Science in Computer Science Machine Learning Concentration
Lodz University of Technology